Does Changing The Wood Of A Solid Body Electric Guitar Change The Sound It Produces?

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1. Introduction

A belief in the electric guitar community says that the wood of a solid body electric guitar changes its tonal characteristics and how it sounds. Being a guitar player myself, I wanted to test out this theory using my physics knowledge and the scientific method. I will be building a guitar rig composed of a plank of wood and a neck to simulate a guitar, I will be recording the sound and analysing it in 5 different ways to find the sound characteristics of each wood to see if there is a relationship between the density of the wood and the: sound duration, loudness, decay, harmonic complexity and tone. All the experiments will be conducted acoustically (the sound is recorded directly from the guitar) and electrically (the sound is amplified with a guitar pickup and amplifier) to see if the wood impacts the sound when the guitar is played both ways.

I personally think that this is only a myth and that the main component to the sound of an electric guitar is the pickup. I hypothesise that there will be little to no change in the sound between the different woods when played electrically but that there will be a difference in the sound between the woods when the guitar is played acoustically because of how resonating bodies work.

2. Sound waves and Resonance

2.1 What a Sound Wave is

A sound wave is a pressure wave which travels through a medium such as air. These mechanical disturbances cause the air molecules to vibrate and the transfer of kinetic energy from particle to particle is in the direction of the wave. This type of wave is called a Longitudinal wave and sound waves are one of the only types of waves observed to behave in this way. The particles of the medium in which the wave is travelling don't move all the way from the source to the detector or target, but only move a small amount causing adjacent particles to move as well thereby repeating the process (What Is a Sound, 2020).

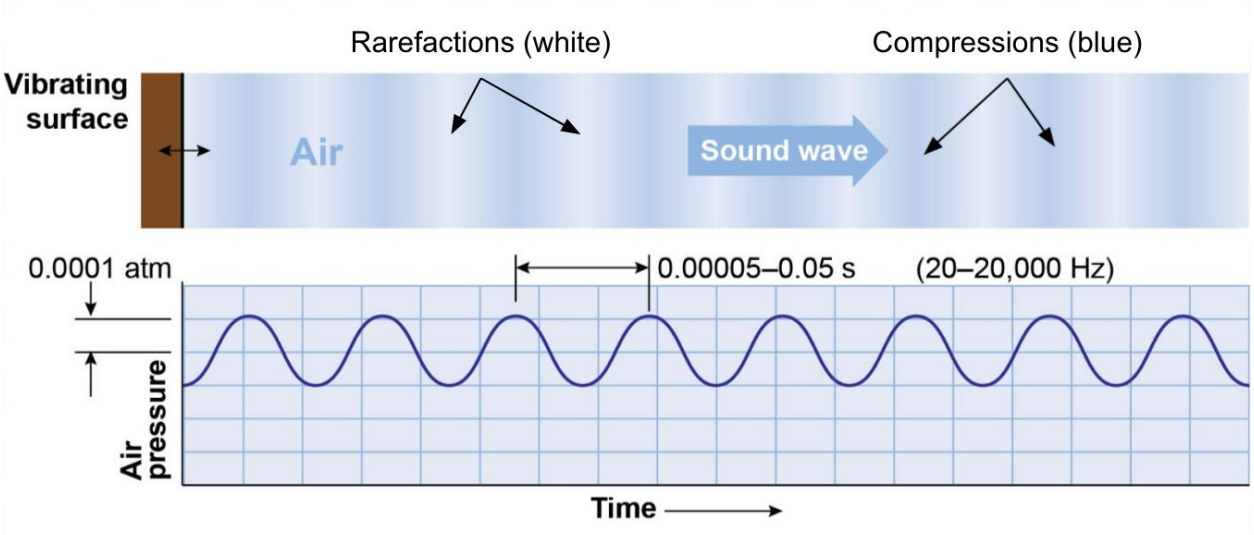


Figure 2.1.1 shows the characteristics of sound wave*s (Diagram of Pressure, 2020).*

The motion of the particles will result in regions of high and low pressure where the particles are compressed and rarefied respectively. Due to the laws of conservation of Momentum and Kinetic Energy, some energy is lost in the collisions between the medium's particles because more particles are encountered as the waves move away from their source. Hence this is why sound does not propagate forever and intensity decreases as you move away from the source. The more distance travelled, the more collisions there are between particles and the more energy is lost (What Is a Sound, 2020).

2.2 Resonance

When an object is left to vibrate, it will do so at a specific frequency unique to that object. That frequency is called the Natural Frequency of an object and it varies with shape, size and composition of the object. An object is said to Resonate when it vibrates at a frequency similar to that of its natural frequency. Therefore, the intensity of and amplitude of these oscillations will be greater (Harris, 2021).



Figure 2.2.1 shows how an acoustic guitar resonates at different frequencies (Resonance on an Acoustic, n.d.).

Sand has been placed on the top of this acoustic guitar. The patterns seen are a result of the vibrations of the top when a certain frequency is played. The different parts of the body of an acoustic guitar resonate differently because of their shape and position on the guitar. Acoustic guitar tops have strips of wood glued onto the back of the top in specific places so that the top vibrates and resonates when a note is played. This is done so that the sound is amplified.

2.3 Harmonics

If you go onto an online tone generator which generates a sound according to the input frequency, you'll notice that the sound coming out of your speakers will be much different to the sound that comes from an instrument playing the same note. This is because the tone generator lacks in harmonics and is only playing the fundamental frequency of the note.

Harmonics are responsible for the unique sound characteristics of different instruments. When a string vibrates, not only does it vibrate from end to end, but there are also smaller vibrations within the string. The frequency of these small vibrations is a multiple of the frequency of the vibration of the whole string, its fundamental frequency (*Harmonic Series*, 2021). Where these vibrations are and how many of them are present when the string instrument is played gives it its Harmonic characteristic. One reason for why some guitars sound better than others might be because the sound coming out of the instrument is more harmonically complex and the sound has more overtones that make the note richer and more interesting.

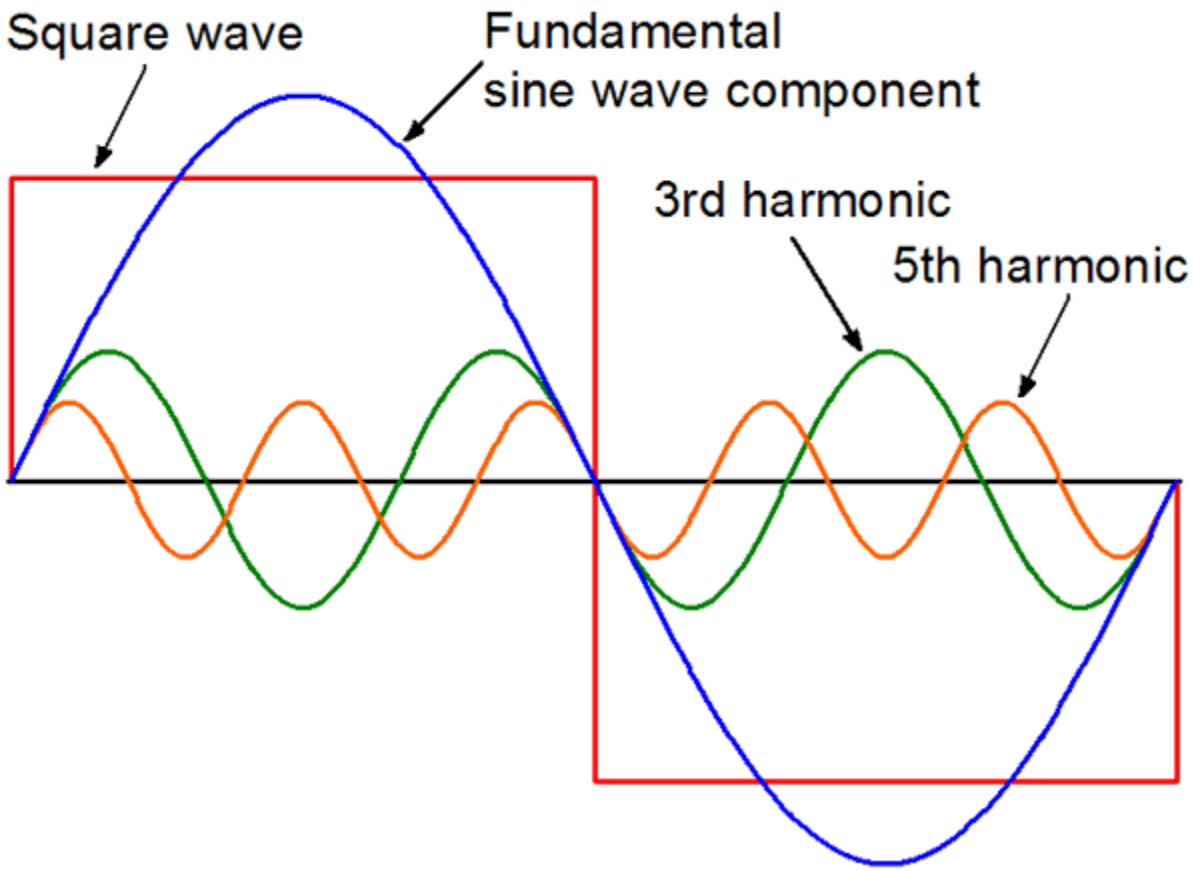


Figure 2.3.1 showing a sine wave and its harmonics (Diagram of the Harmonics, n.d.).

3. How an Electric Guitar Works

The magnetic pickup is the heart of an electric guitar, it is what gives the guitar its distinct sound. The 6 cylindrical magnets in the pickup produce a magnetic field which is disturbed when the steel guitar strings link the magnetic flux which induces an EMF in the electronics of the guitar (Lawing, 2018). This EMF is carried to the amplifier via the jack cable and the amplifier turns the electrical signal into sound by vibrating the cone inside of its speaker. These vibrations create the high and low pressure waves by pushing the air molecules around which we then perceive as sound (Seydel, 2017).

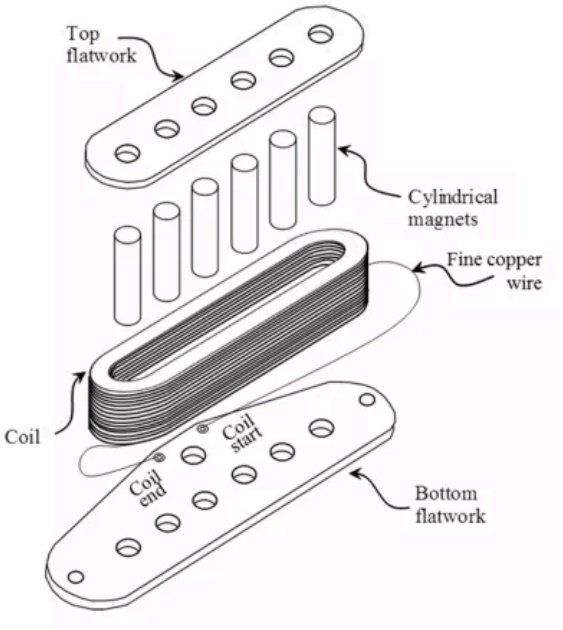


Figure 3.1.1 shows the construction of a standard Single Coil pickup (Construction of a Single, 2018).

4. Methodology / Experimental Description

4.1 Guitar Rig

To accurately simulate an electric guitar, an experimental guitar rig was constructed. It consisted of a plank of wood (the independent variable) as the body, a block of wood used to hold a guitar tuning machine (used to maintain tension on the string so that the note played was consistent) and a tailpiece to hold the other end of the string. The guitar was tuned to an E4 (the 4th E note on a Piano Keyboard) at 329.63 Hz (Szynalski, n.d.). A guitar pickup wired directly to the output jack and was screwed under the string so that experiments could be conducted both acoustically and electrically. The wooden

plank constituted the majority of the mass of the guitar rig so it was the main resonating body. The pieces of wood were cut by professionals using a machine so the degree of error was minimal. The same neck, pickup, tailpiece, tuner, guitar string and screws were used to build the 7 different guitar rigs to keep consistency. Measurements for the placement of each component were measured to the nearest millimeter.

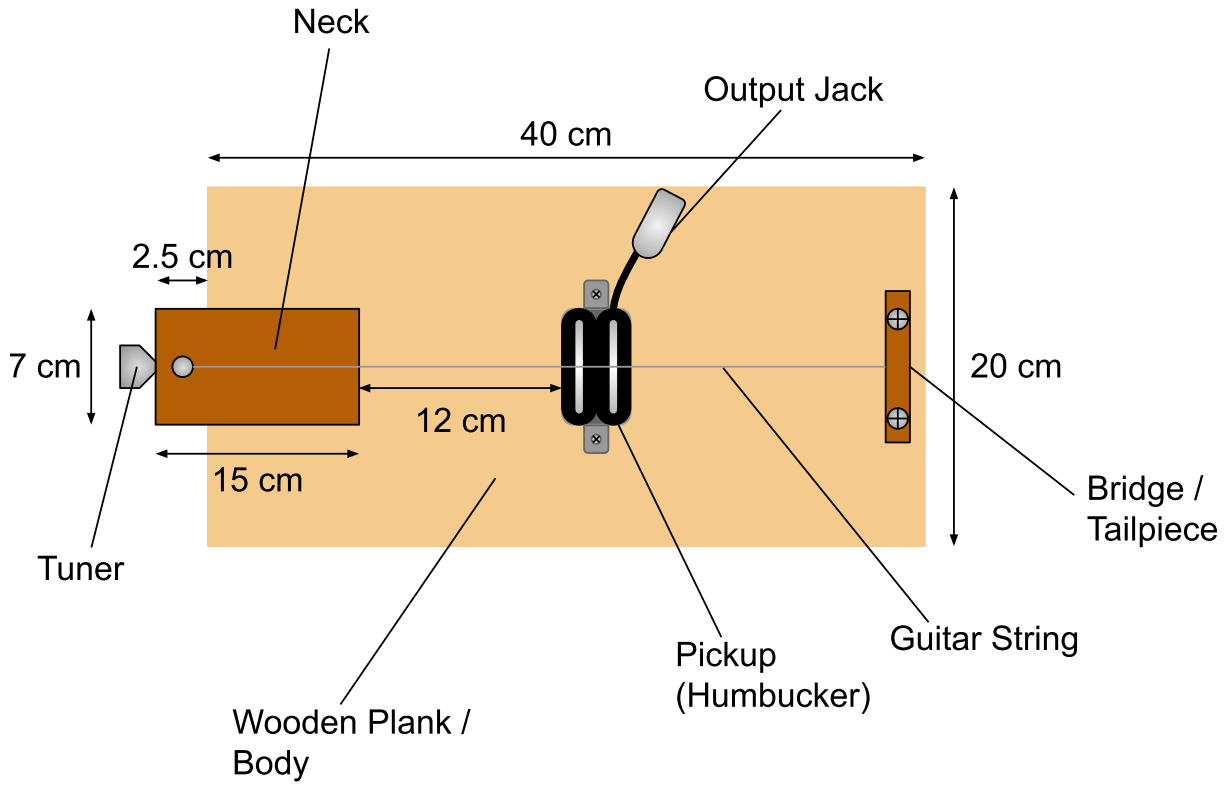


Figure 4.1.1 shows a diagram of the guitar rig.

4.2 Experimental Method - Sound sample capture

A multitude of experiments will be conducted and the data will be analyzed in four different ways. But the capture of the sound samples will be similar for all of the experiments.

The sound capture for the acoustic samples will be conducted by holding the guitar rig on my lap in the standard guitar playing position. The microphone will be held by a stand 1 cm above the pickup with the gain at 100%. The gain control serves the purpose of increasing the amplitude of the waves detected by the microphone so the signal is stronger and easier to work with, the percentage tells the user how much this function is engaged (100% being the maximum). The sound capture for the electric

samples will be conducted by micing up the Amplifier with the microphone at the center of the speaker cone, two finger widths from the front. Gain will be at 100% for the first set of experiments to observe the sound sustain difference between the woods, the second set will have the gain at 50% to observe the volume difference between the woods. The microphone will be plugged into the computer via a sound interface that converts the signals from the microphone into something the computer can understand.

Each experiment will consist of 5 trials, picking attack consistency will be maintained by always plucking the string in the same way and seeing if the waveform is consistent and if the trials sound similar to each other, if a mistake is made (ex: pick attack was too violent) the trial will be redone.

4.3 Experimental Method - LogicProX and Matlab application / use

I will be using two softwares for my sound analysis: LogicProX will be used for density v/s sound duration, volume and sound decay which are experiments 1, 2 and 3 respectively. Matlab will be used for the Fourier Transform analysis (experiment 4).

For experiment 1, the length of the note will be measured using a noise gate. A noise gate cuts the signal when it falls below a certain level to indicate when the note is over. It will be set at -50 dB (LogicProX scale). The LogicProX scale indicates how much a sound is below its original volume, 0 being how loud the sound originally was and -50 dB for example indicates that the sound has been turned down by 50 decibels. LogicProX also displays the volume level of the track in real time, which will be used to measure the peak volume (experiment 2) and to plot a sound decay diagram for experiment 3.

In Matlab, the sound will be recorded using a sample rate of 8000 samples per second and a duration of one second. This means that the highest frequency that can be detected after the Fourier Transform is 4000 Hz which is more than enough since the note that I'm playing has a natural frequency of 329.63 Hz. The data points will then undergo a Fourier Transform so that the sound wave can be deconstructed into its main components to see what the components of the note are (harmonic complexity).

All experiments will be conducted with both the acoustic and electric sound samples to see if there is a difference.

4.4 How a Microphone works

A microphone picks up the vibrations in the air created by sound waves. The high and low pressure waves make a small diaphragm, similar to that of a loudspeaker, vibrate back and forth inside of the microphone. A coil is attached to the diaphragm and it moves along with it. Inside the coil is a stationary magnet which creates a magnetic field. The way in which a microphone detects sound is similar to how a magnetic pickup detects a sound. The coil produces disturbances within the magnet's magnetic field and an electrical signal comes out which is then transformed into sound (Mitchell, n.d.). For this experiment I have a Shure SM58 which is an industry standard high quality microphone.

4.5 The Fourier Transform

The Fourier Transform can be described as an inverse to wave superposition. It can take any wave function and break it down into its components and produce a graph showing what frequency those components have (3Blue1Brown, 2018). Below is one of the Fourier Transform graphs from my experiments.

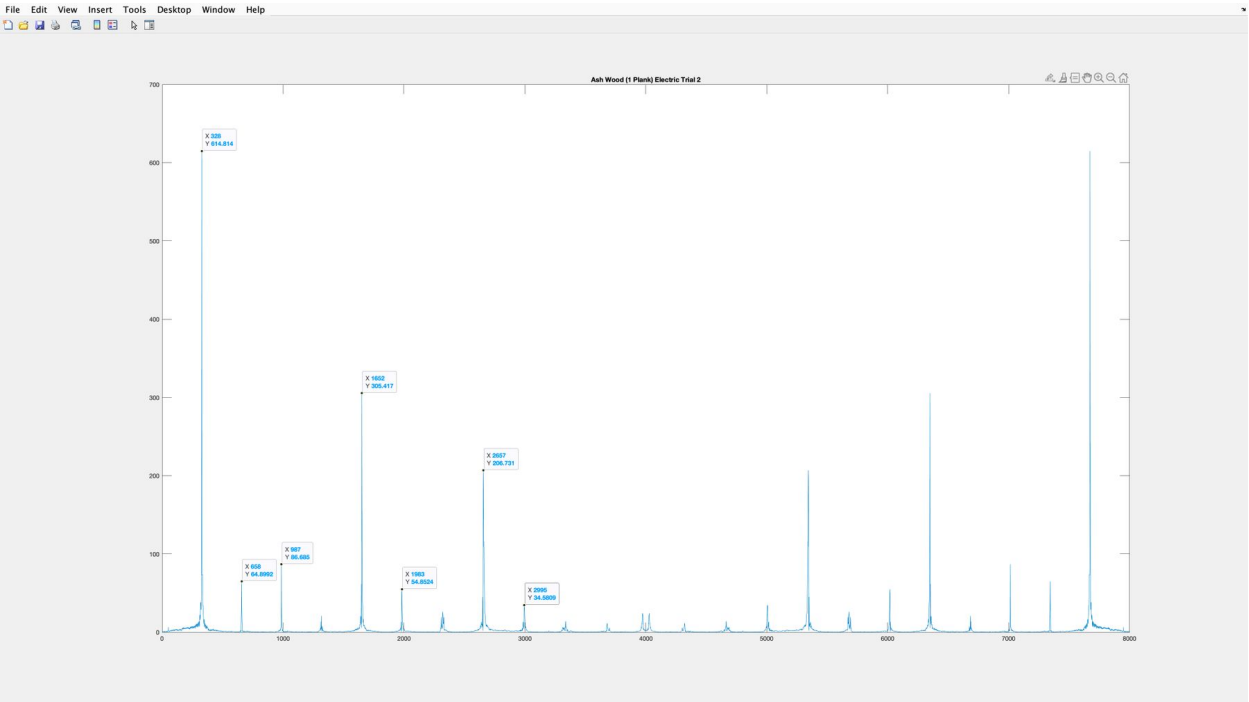


Figure 4.5.1 shows the Fourier Transform graph for Ash wood Trial 2 (Electric Experiment)

Due to the fact that a Fourier Transform outputs imaginary numbers, the absolute value of x was graphed making the graph symmetrical. The x-axis represents the frequency and the y-axis represents the intensity. The peaks shown are the different component

frequencies that make up that sound wave. The higher the peak the louder it was, so the tallest peaks are the main component frequencies. The tallest “spike” is the dominant frequency, E4 in this case. The more spikes a graph has, the richter a sound is which could be a desirable quality. The mathematics behind the Fourier Transform are very interesting but go beyond the scope of this essay.

4.6 Different woods tested

In these experiments, 7 different woods were tested. The wooden planks were cut to the right dimensions by a professional using a factory machine. Figure 3.4.1 below shows the types of woods used and their properties.

Wood Type / Species	Height ± 0.05 cm	Length ± 0.05 cm	Width ± 0.05 cm	**Volume cm^3**	% ∆V	Unc V cm^3	**Mass ± 25g**	% ∆M	Unc M g	**Density g/cm^3**	Unc D g/cm^3	% ∆D	**Reference Density Value (g/cm3)**
Pine	1.85	40.15	20.0	**1486**	3.1	50	**650**	3.8	25	**0.44**	0.3	6.9	**0.42**
Acacia	1.80	40.10	20.0	**1444**	3.2	40	**725**	3.4	25	**0.50**	0.3	6.6	**0.69 - 0.88**
Cherry	1.75	38.80	19.9	**1351**	3.2	40	**750**	3.3	25	**0.56**	0.4	6.6	**0.63**
Ash	1.80	40.00	19.9	**1433**	3.2	40	**825**	3.0	25	**0.58**	0.4	6.2	**0.54**
Lime	1.85	40.00	19.9	**1473**	3.1	40	**975**	2.6	25	**0.66**	0.4	5.6	**0.56**
Oak	1.90	40.00	20.0	**1520**	3.0	50	**1050**	2.4	25	**0.69**	0.4	5.4	**0.6 - 0.9**
Chipboard	1.60	40.10	20.0	**1283**	3.5	40	**900**	2.8	25	**0.70**	0.4	6.3	**No Accepted Value**

Figure 3.4.1 shows the different woods used and their characteristics

Pine, Oak, Cherry, Acacia and Lime were all chosen due to their availability and different density values. Some woods have densities similar to some woods used in the guitar making industry, Oak’s density is similar to Mahogany (0.67 g/cm^3) for example. But Ash is a common wood used to build guitars and Chipboard was chosen to see the difference between reconstructed wood and normal wooden planks.

5. Raw Data for experiments 1, 2 and 3 (from LogicPro X)

In LogicPro X the data was recorded digitally as a waveform.



Figure 5.1 shows a screenshot of the LogicPro X window (Electric experiment Oak Trial 1, Gain at 50%)

The sound then can be manipulated and observed using the built in functions of the software. The noise gate can be set so that the sound gets cut off after it falls under a certain noise level, and the volume level is shown above the volume slider (-7.6).



Figure 5.2 shows the noise-gate window



Figure 5.3 shows the volume slider window

6. Processed Data For Experiments 1, 2 and 3 (from LogicPro X)

6.1 Experiment 1: Density v/s Sound Duration

To measure the sound duration, the noise gate was engaged at -50 dB (LogicProX scale) and the time recorded at the shut off of the sound indicates the sound duration. The loudest volume level is shown above the volume slider and it was recorded. Below is the data sample for Oak (Acoustic trials).

Acoustic Trials -- Noise Gate Engaged at: -50 dB -- Microphone gain at 100%						
Wood Type -- Density	Oak Wood -- 0.69 g/cm3					
Trial	1	2	3	4	5	**Average**
Peak Volume (dB)	-5.2	-5.3	-2.8	-4.6	-3.4	**-4.3**
Duration (s) ± 0.2s	3.2	3.2	3.0	3.0	2.8	**3.1**

Figure 6.1.1 shows a sample data set from the sound duration experiment

The average Sound Durations for each experiment were recorded on this data table for better clarity, and the data table was used to construct the graph below. This was done also for the Peak Volume data.

Wood	**Density (g/cm3)**	**Sound Duration (Acoustic) (s)**	**Sound Duration (Electric 50%) (s)**	**Sound Duration (Electric 100%) (s)**
Oak	0.69	3.10	2.50	4.70
Pine	0.44	4.30	4.50	8.50
Acacia	0.50	4.30	4.60	9.40
Ash	0.58	3.80	2.50	6.20
Cherry	0.56	3.00	2.20	5.80
Lime	0.66	4.50	3.50	6.90
Chipboard	0.70	5.20	5.40	11.70

Figure 6.1.2 shows the sound duration data for all the woods

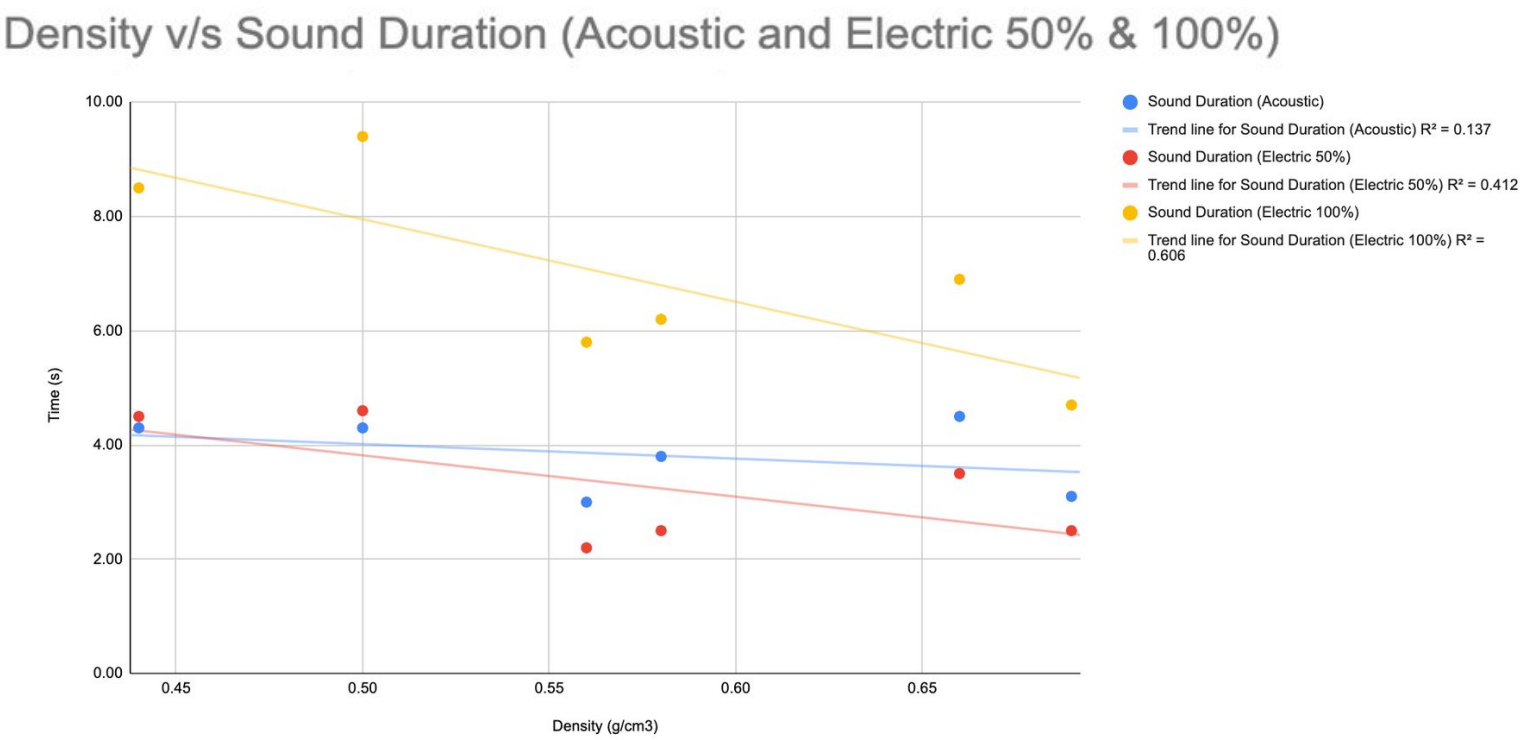


Figure 6.1.3 shows the data from the table plotted on a scatter graph

Density v/s Sound Duration presented a weak correlation for the acoustic trials and a slightly stronger correlation with the electric trials. The correlations observed might be due to randomness; it would have made sense for the acoustic experiment to have a stronger correlation since the sound is made by the string vibrating and marginally resonating with the wooden plank. But the strongest correlation is with the electric experiments who's sound mainly comes from the pickup which was kept constant. A possible explanation could be that the string wasn't plucked consistently. The data points on the graph also do not form a line and the high R^2 number might be due to the distance between the points above and the points below averaging out.

6.2 Experiment 2: Density v/s Volume

A graph for wood density v/s peak recorded volume was also constructed, but for both electric and acoustic trials, no correlation was observed. Since this is a solid body electric guitar, the pickup is the main component of the sound. But even when unplugged, the sound doesn't get amplified by the solid wooden plank because the

sound doesn't resonate as much inside of it. Similar to the Harmonic Complexity experiment, if this were done with an acoustic guitar, a correlation might be observed.

6.3 Experiment 3: Density v/s Sound Decay

To construct a sound decay graph for the recorded waveforms, the volume level was taken at time intervals of 0.5 seconds and recorded in the table below. A table was constructed for both acoustic and electric experiments.

Time (s)	Volume (Acoustic Trial)						
	Pine	**Acacia**	**Cherry**	**Ash**	**Lime**	**Oak**	**Chipboard**
0.0	-8.1	-13.9	-13.4	-10.8	-11.4	-5.3	-5.4
0.5	-27.5	-28.3	-24.5	-23.3	-26.3	-23.5	-23.3
1.0	-28.0	-32.3	-33.0	-35.3	-27.3	-32.9	-27.3
1.5	-33.1	-40.3	-38.6	-36.1	-35.5	-36.1	-29.2
2.0	-36.8	-42.4	-45.2	-40.4	-36.1	-42.2	-32.8
2.5	-41.0	-43.0	-50.0	-43.0	-42.5	-43.2	-35.5
3.0	-42.4	-45.8	-54.8	-46.5	-42.5	-46.4	-36.5
3.5	-45.1	-49.9	-57.3	-49.8	-46.0	-46.8	-39.6
4.0	-46.9	-52.9	-56.7	-51.0	-46.0	-49.0	-41.0
4.5	-48.2	-51.6	-57.1	-54.3	-52.3	-49.2	-42.2
5.0	-49.2	-53.6	-56.4	-54.6	-53.3	-50.3	-43.2
5.5	-50.6	-55.9	-57.9	-55.0	-53.3	-51.0	-44.8
6.0	-50.0	-56.3	-57.7	-56.0	-54.7	-51.4	-45.9
6.5	-51.0	-56.7	-58.6	-57.0	-56.4	-50.4	-46.2
7.0	-52.3	-56.9	-58.2	-55.6	-57.8	-50.4	-43.2
7.5	-51.0	-56.8	-58.1	-55.8	-58.0	-50.9	-47.8
8.0	-52.0	-58.3	-59.3	-57.3	-57.4	-51.2	-50.0
8.5	-52.0	-57.8	-56.6	-57.9	-57.5	-52.0	-51.2
9.0	-51.4	-57.6	-56.6	-58.2	-57.7	-53.3	-50.5

Figure 6.3.1 shows the sound decay data table

The data table was used to construct a sound decay graph, the sound decay graph for the Acoustic trial is shown below.

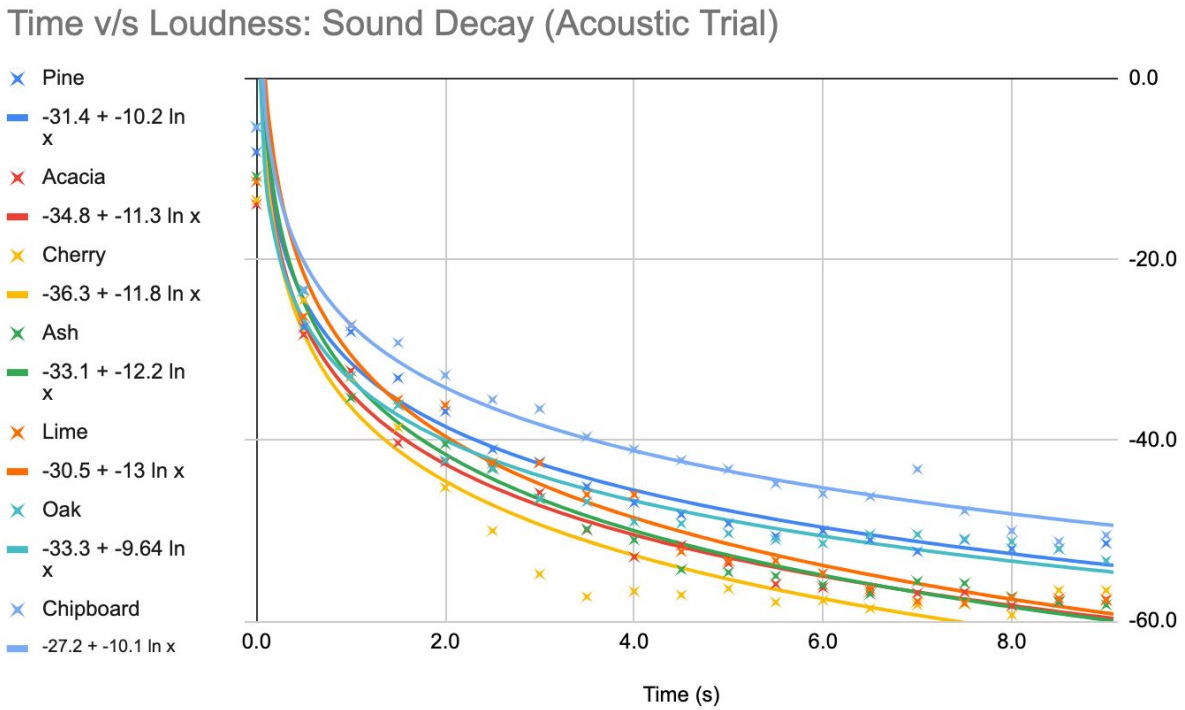


Figure 6.3.2 shows the sound decay graph of each wood

Density v/s Sound Decay was the only quantitative data set that presented a correlation. The decay constant (the number multiplied by ln) was taken for each wood and a data table and graph were constructed per experiment for a total of 3 data tables and 3 graphs.

Wood	Density	Decay Constant
Pine	0.44	-10.2
Acacia	0.50	-11.3
Cherry	0.56	-11.8
Ash	0.58	-12.2
Lime	0.66	-13.0
Oak	0.69	-9.6
Chipboard	0.70	-10.1

Figure 6.3.3 shows the sound decay constant from the sound decay graph for each wood

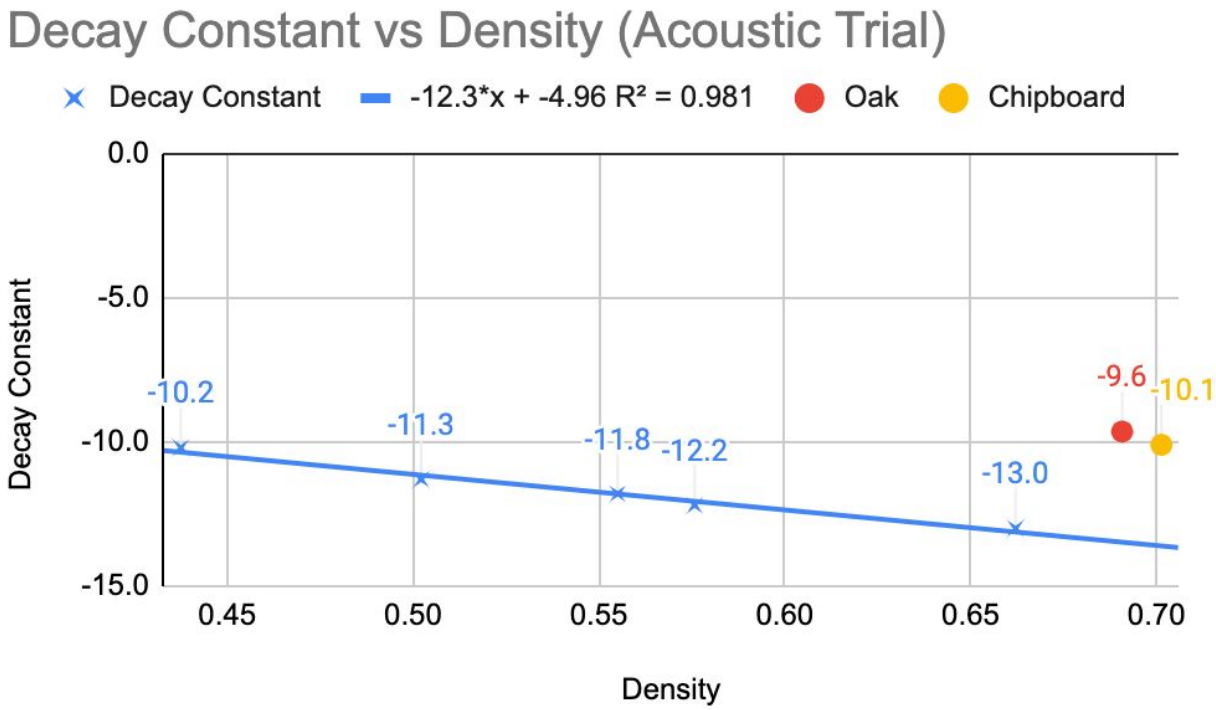
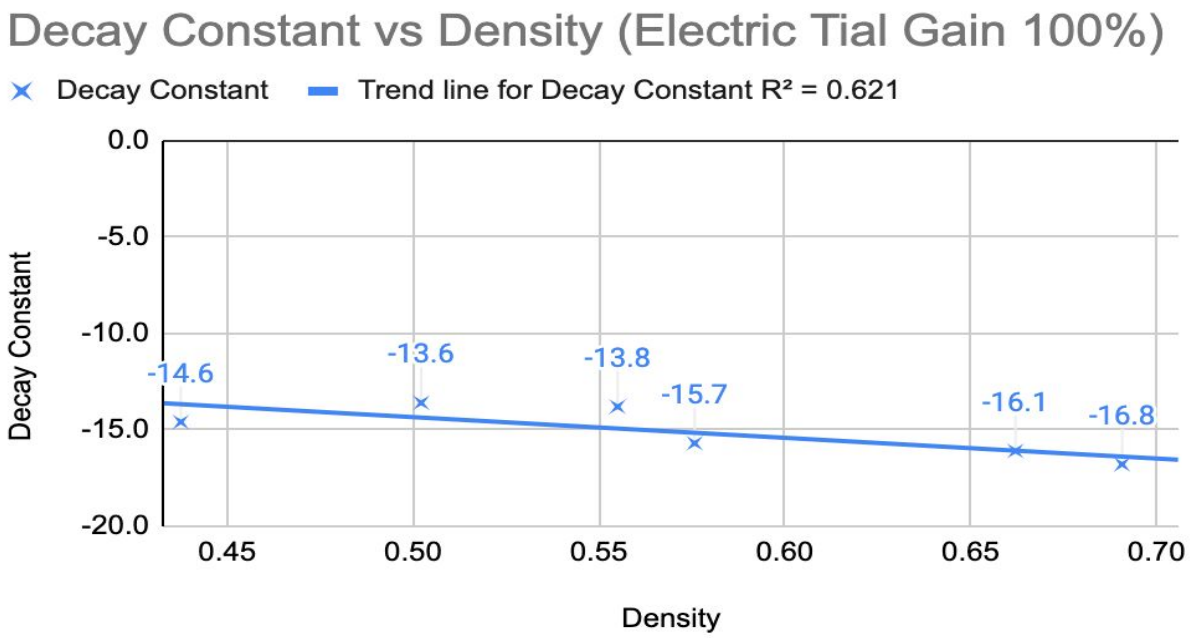
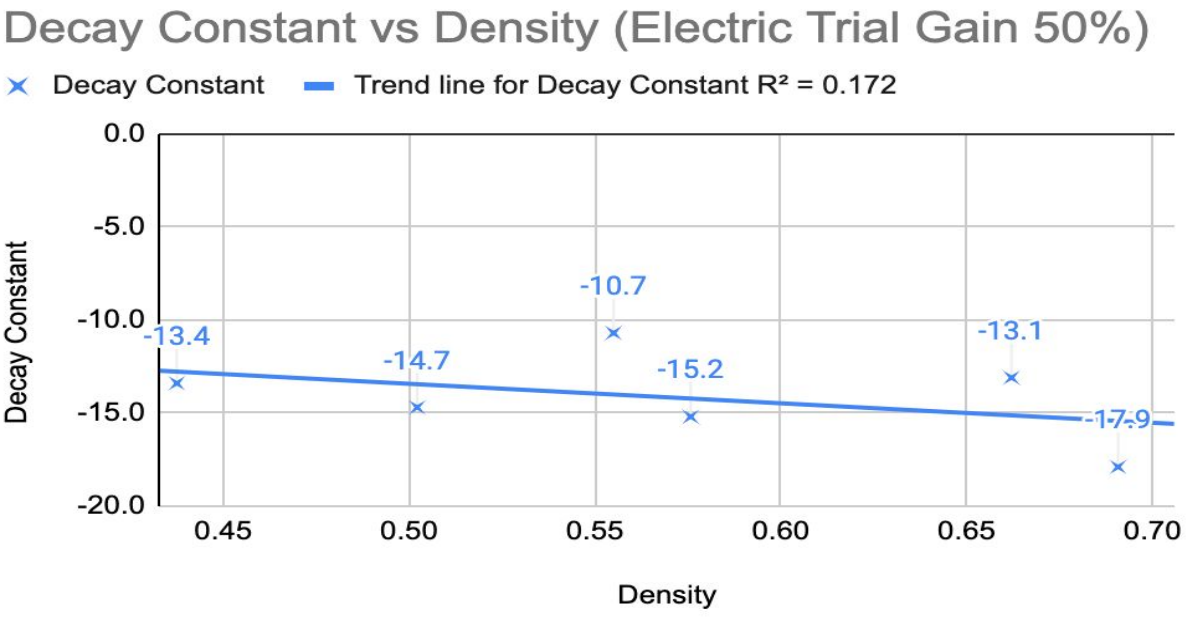


Figure 6.3.4 shows the data from the table plotted on a scatter graph

Chipboard and Oak were outliers but the rest of the data points present a strong correlation between the wood density and the sound decay constant. Chipboard was tested to see the difference between a recomposed wood and a normal wooden plank, but the fact that Oak is an outlier may be due to human error or another unknown factor.

The decay constant represents how fast the sound decays, the higher the number the slower the sound decays. Resonance between the wooden plank and the string is minimal because the plank is solid and not hollow, but a correlation is present. This may be due to the wood strength. In the electric experiments, a weak correlation was found for the Gain at 100% experiments.





Figures 6.3.5 and 6.3.6 show Wood density v/s Sound decay constants from the data for the electric trials at 100% gain and 50% gain

Chipboard was taken out of the data sets due to its nature as a recomposed wood. The fact that this relationship is weaker than the one for the acoustic experiment and that the electric experiments with Gain at 50% presented no correlation at all would support the hypothesis that the wood has little to no impact on the sound sustain of an electric guitar due to the pickup being the what captures the string vibrations and turns it into sound.

7. Raw Data For Experiment 4 (from Matlab)

In Matlab the sound was recorded by measuring the air pressure at small time intervals using the microphone. The data was displayed as a list of data points which then underwent a Fourier transformation and were plotted on a graph.

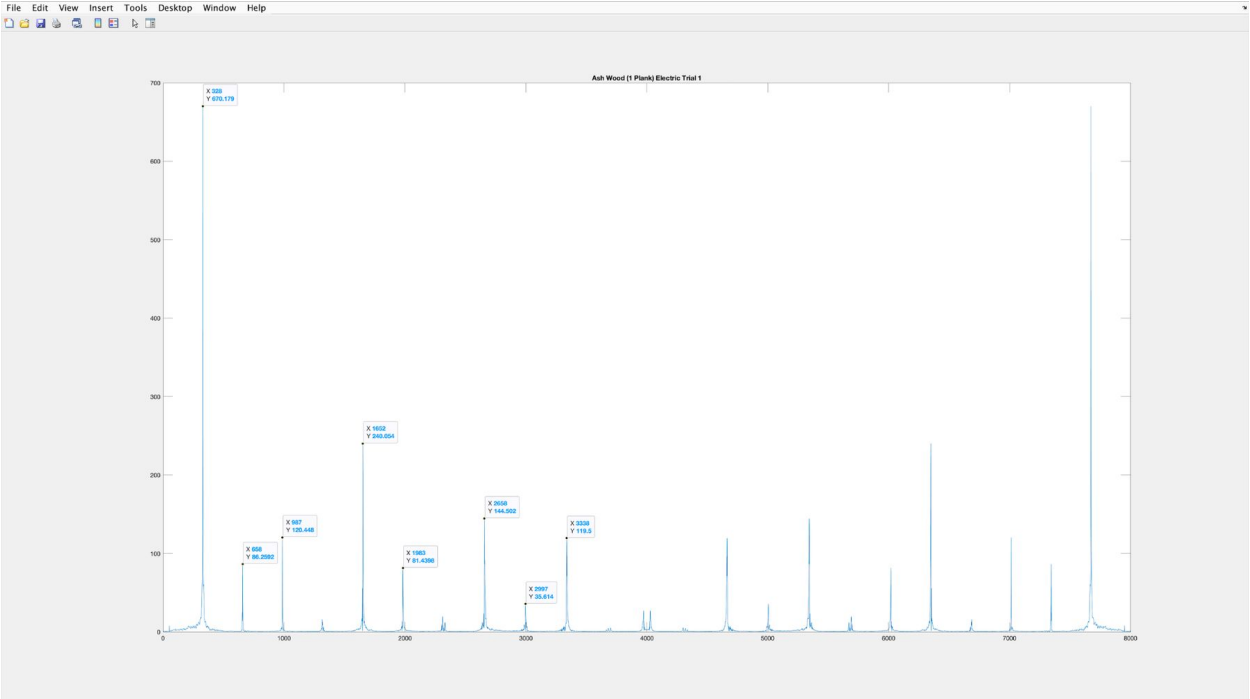


Figure 7.1 shows a Fourier Transform graph from Matlab

Due to the fact that a Fourier Transform outputs imaginary numbers, the absolute value of x was graphed making the graph symmetrical. The x-axis represents the frequency and the y-axis represents the intensity. The tallest “spike” is the dominant frequency, E4 in this case.

8. Processed Data For Experiment 4 (from Matlab)

8.1 Harmonic complexity

The frequency and quantity of the peaks of the Fourier Transform graphs were recorded in this data table.

Wood	Trial	Number of Peaks		Dominant Frequency (Hz)		Dominant Note	
		Acoustic	Electric	Acoustic	Electric	Acoustic	Electric
Oak (1 Plank)	1	2	4	326	325	E4	E4
	2	2	3	325	326	E4	E4
	3	2	3	325	326	E4	E4
	4	2	4	325	326	E4	E4
	5	2	3	325	326	E4	E4
	Average	**2**	**3.4**	**325.2**	**325.8**	**E4**	**E4**
Oak (2 Planks)	1	5	1	660	329	E5	E4
	2	4	1	659	329	E5	E4
	3	3	1	659	329	E5	E4
	4	3	1	659	329	E5	E4
	5	5	1	659	329	E5	E4
	Average	**4**	**1**	**659.2**	**329**	**E5**	**E4**
Pine (1 Plank)	1	2	2	330	327	E4	E4
	2	2	1	330	328	E4	E4
	3	2	1	330	328	E4	E4
	4	6	2	662	328	E4	E5
	5	5	2	662	328	E4	E5
	Average	**3.4**	**1.6**	**330**	**327.8**	**E4**	**E4**
Chipboard (1 Plank)	1	3	3	329	323	E4	E4
	2	2	2	328	328	E4	E4
	3	4	4	657	327	E5	E4
	4	3	4	327	327	E4	E4
	5	4	3	326	328	E4	E4
	Average	**3.2**	**3.2**	**327.5**	**326.6**	**E4**	**E4**

Figure 8.1.1 showing the Fourier Transform graph data

For both the electric and acoustic experiments, no correlation was observed between the number of peaks in the Fourier Transform graph and the density of the wood. When a musical note is played by an instrument it doesn't only play the frequency of the note that is played, but the sound is a sum of many different frequencies which give the instrument its unique sound characteristics and harmonic complexity. By counting the number of peaks in a Fourier Transform graph, we can see how many components the note has otherwise known as the note's overtones. A richer and more complex sound may be preferable because it gives the instrument character, something that would otherwise be absent if the only frequency played was the frequency of the desired note.

The difference in the number of peaks between the different graphs may be due to other characteristics of the wood such as strength or amount of grain present or just randomness / human error. Overtones are created when an oscillating resonator (in this case a string) oscillates in many different modes. These modes have different frequencies and when summed together create the note we hear. The number and frequency of the modes depends on a resonating body. Since the wooden plank is solid, the only two things resonating are the string and the column of air. An acoustic guitar is louder than an unplugged solid body electric guitar because the body resonates and amplifies the sound. If this experiment was conducted with acoustic guitars with different woods, a correlation might have been observed.

9. Qualitative Data

Even if the main component of the sound coming out of an electric guitar is the pickup, there was an audible sound difference between the different woods and each had their own characteristics.

Wood	Sound description (the Tone)	
	Acoustic	Electric
Pine	Harsh attack with a thin and “glassy” tone. Long sustain which cyclically increases and decreases in volume.	The cyclical volume changes are very evident and have a similar time period to the ones in the acoustic trials. For the rest it still has a glassy tone with a harsh attack and by far the longest sustain.
Acacia	Softer attack than Pine and a rounder tone. An overtone can be heard playing a very high pitched note. Sound decays constantly without the cyclical increases and decreases in volume that were heard with Pine.	Softer attack than Pine but has a round tone, unlike the acoustic trials, these trials don’t have the high pitched overone.
Cherry	Round tone with a softer attack than Acacia. The sound is almost “mellow” after the initial harsh attack due to the pick plucking the string.	Mellow tone with a soft attack but it “opens up” as the sound fades out.
Ash	An open sound less mellow than Cherry but with an even softer attack. Has the cyclical volume changes that Pine has but with a shorter period.	Harsher tone than Cherry with a strong attack, this sound is similar to Pine but lacks the sustain.
Lime	Very closed and mellow tone, similar to the neck pickup sound in an electric guitar. Softer attack and shorter sustain.	Very closed and mellow tone. The sound reflects the acoustic trials.
Oak	Very glassy tone with a harsh attack. It is similar to Pine but more aggressive. Similar sustain to Pine.	Glassy tone which dies out quickly. You can hear an overtone playing a very high pitched note.
Chipboard	Similar glassy tone to Oak and Pine with a moderate sustain. But has a mellower tone than Oak and Pine.	Harsh and glassy tone similar to Oak but has a much greater sustain.

Figure 9.1 shows the qualitative data from the various experiments

Apart from a few outliers, a few patterns can be observed: as density increased the sound became more mellow and rounder, but it lost sustain. The sharpness of the attack also decreased as density increased. Overall which wood is the best sounding is up to personal preference, to me Ash sounded best acoustically, electrically Ash and Cherry were my favorite but the unique sustain that Pine makes it very interesting and gives it a unique sound. There was a more audible change in the sound between the different woods with the acoustic trials more so than the electric trials. But there are notable differences in the sound produced by each wood, both in the acoustic and electric trials.

10. Conclusion

Experiments 1 and 2 presented no correlation for both the acoustic and electric trials which shows that the density of the wood doesn't affect the sustain nor the volume of the sound coming out of the guitar. Experiment 3 on the other hand, presented a very strong correlation between Density and Sound Decay for the acoustic trials, less so for the electric trials. The Fourier Transform analysis (Experiment 4) showed no correlation between Density and Harmonic Complexity for both the acoustic and electric trials.

Overall the acoustic trials' results showed to be more affected by the change in wood density than the electric trials. The sound change from one wood to another, as stated in the qualitative data analysis, was more evident in the acoustic trials. This shows that the wood density has an effect on the sound of a solid body electric guitar when played acoustically, less so electrically. This makes sense because the wooden plank is what resonates and amplifies the sound when the guitar is played acoustically. So the sound will obviously change if there is a change in the wooden plank. But if the guitar is played electrically, the vibrations of the strings are picked up by the pickup which turns the disturbances in the magnetic field into sound through an amplifier. So the wood has nothing to do with the sound coming out of the guitar.

In conclusion, the belief that the wood affects the tone and sound properties of the electric guitar is partly false. There were some audible differences between the samples of the different woods when the guitar rig was played electrically, but there is no change in sustain, volume or harmonic complexity. Note that my guitar rig was very barebones

and a real electric guitar is bigger and composed of many different parts. The wood that makes up the body is only one part and there are many different factors which could affect the sound of an electric guitar.

11. Strengths and Weaknesses of the Experiment

This experiment is very simple and “bare bones”, there aren’t many steps which could result in high levels of uncertainty. It also reflects how a guitar works in real life. The only way this could be improved is to build 7 fully functioning guitars instead of smaller guitar rigs, but that would be costly and time consuming. The values / readings are all taken by a machine and analysed using software meaning that uncertainties are negligible. The only uncertain value in this experiment is the value for the density of the wood, a value which can be found and checked on the internet.

But this experiment does have its weaknesses. There is a very large human component, the guitar rig has to be built by hand which could lead to inconsistencies but this can be fixed by simply being very careful and verifying that no mistakes have been made while building the guitar rig. The guitar is also played by a human and pick attack and speed can change between trials. One can fix this by either doing many trials which could be very time consuming, or one could build a machine that plays the guitar for you. But this can prove to be very difficult having done it myself unsuccessfully. But overall this experiment is well designed and appropriate for the research that has been conducted.

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